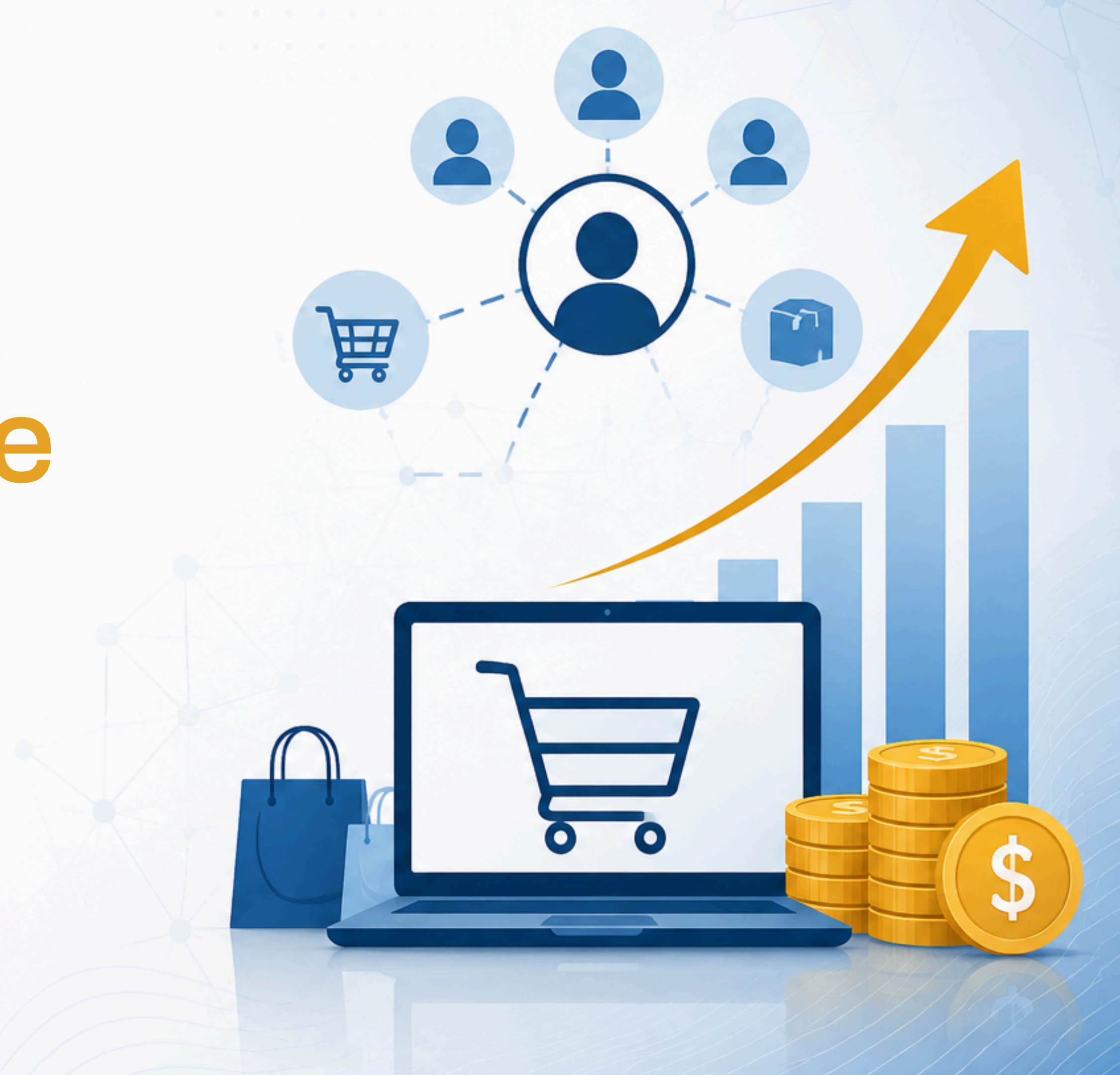


Lab Exam Part 3

Customer Lifetime Value Prediction

Probabilistic BTYD and Machine Learning Approach

Irene Marrali
AI for Communication and Marketing



Macro View: Financial Planning

BTYD - Best for Financial Planning

Lower RMSE indicates better control of **large forecasting errors**, especially for high-value customers.

Recommended for customer-base valuation and quarterly budget allocation.

XGBoost - Best for Individual Prediction

Lower MAE indicates more accurate revenue predictions for the **typical customer**.

Recommended for customer-level targeting and operational marketing decisions.



Model	MAE	RMSE
BTYD	11.75	36.31
XGBoost	5.71	39.48

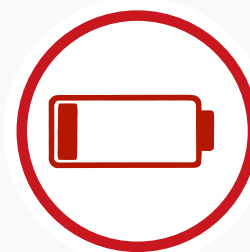
Micro View: Operational Marketing

Who should be prioritized for a win-back campaign?



10

Priority Win-Back Customers



6%

Average Probability Alive



\$135.73

Total Predicted 90-Day CLTV

Recommended Action

Launch a **highly targeted, low-cost reactivation campaign** using personalized reminders, limited-time incentives and product recommendations based on previous purchases.